SF1811 Optimization
A note on linear programming optimality and duality

In the lectures, duality and optimality conditions are first derived for a linear program in standard form. This gives the primal-dual pair of linear programs

$$(P_{\text{LP}}) \quad \begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b, \quad x \geq 0, \end{array} \qquad (D_{\text{LP}}) \quad \begin{array}{ll} \text{maximize} & b^T y \\ \text{subject to} & A^T y + r = c, \quad r \geq 0. \end{array}$$

Then, $\hat{x}$ is optimal for (P_{LP}) and $(\hat{y}, \hat{r})$ is optimal for (D_{LP}) if and only if $\hat{x}$ is feasible to (P_{LP}), $(\hat{y}, \hat{r})$ are feasible to (D_{LP}), and $\hat{x}_j \hat{r}_j = 0$, $j = 1, \ldots, n$. (Note that as $\hat{x} \geq 0$ and $\hat{r} \geq 0$, we have $\hat{x}_j \hat{r}_j = 0$, $j = 1, \ldots, n$, if and only if $\hat{x}^T \hat{r} = 0$.)

This formulation is useful because it uses the same quantities as in the derivation of the simplex method. At every iteration, the simplex method maintains primal feasibility,

$$Ax = b, \quad x \geq 0,$$

as well as

$$A^T y + r = c, \quad x^T r = 0.$$

When $r \geq 0$, dual feasibility also holds. All primal-dual optimality conditions are then satisfied, and the method terminates.

The purpose of this note is to show that the results for the general form, as stated on the formula sheet, follow from the results for standard form derived in the lectures. Different forms look different but contain equivalent information. The formula sheet gives a primal-dual pair and corresponding optimality conditions for a linear program in general form:

- **Optimality conditions for LP problem in general form.**

$$(P) \quad \begin{array}{ll} \text{minimize} & c_1^T x_1 + c_2^T x_2 \\ \text{subject to} & A_{11} x_1 + A_{12} x_2 \geq b_1, \\ & A_{21} x_1 + A_{22} x_2 = b_2, \\ & x_1 \geq 0, \ x_2 \text{ free}. \end{array} \qquad (D) \quad \begin{array}{ll} \text{maximize} & b_1^T y_1 + b_2^T y_2 \\ \text{subject to} & A_{11}^T y_1 + A_{21}^T y_2 \leq c_1, \\ & A_{12}^T y_1 + A_{22}^T y_2 = c_2, \\ & y_1 \geq 0, \ y_2 \text{ free}. \end{array}$$

  $\hat{x}$ is optimal for (P) and $\hat{y}$ is optimal for (D) if and only if $\hat{x}$ is feasible to (P), $\hat{y}$ is feasible to (D), $\hat{y}_1^T (A_{11} \hat{x}_1 + A_{12} \hat{x}_2 - b_1) = 0$ and $\hat{x}_1^T (c_1 - A_{11}^T \hat{y}_1 - A_{21}^T \hat{y}_2) = 0$.

  ...

In order to show that the pair (P) and (D) and corresponding optimality conditions follow from those of the pair (P_{LP}) and (D_{LP}), we may put (P) into standard form by letting $x_2 = x_2^+ - x_2^-$, with $x_2^+ \geq 0$, $x_2^- \geq 0$, and adding a nonnegative slack variable x_3 to the first constraint, which gives the equivalent problem

$$(P') \quad \begin{array}{ll} \text{minimize} & c_1^T x_1 + c_2^T (x_2^+ - x_2^-) \\ \text{subject to} & A_{11} x_1 + A_{12} (x_2^+ - x_2^-) - x_3 = b_1, \\ & A_{21} x_1 + A_{22} (x_2^+ - x_2^-) = b_2, \\ & x_1 \geq 0, \ x_2^+ \geq 0, \ x_2^- \geq 0, \ x_3 \geq 0. \end{array}$$

Thus, (P') has the standard form (P_{LP}) with

$$A = \begin{pmatrix} A_{11} & A_{12} & -A_{12} & -I \\ A_{21} & A_{22} & -A_{22} & 0 \end{pmatrix}, \quad b = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}, \quad c = \begin{pmatrix} c_1 \\ c_2 \\ -c_2 \\ 0 \end{pmatrix}, \quad \text{and} \quad x = \begin{pmatrix} x_1 \\ x_2^+ \\ x_2^- \\ x_3 \end{pmatrix}.$$

The corresponding dual problem (D_{LP}) then takes the form

$$\begin{aligned}
&\text{maximize} && \begin{pmatrix} b_1^T & b_2^T \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \\
&\text{subject to} && \begin{pmatrix} A_{11}^T & A_{21}^T \\ A_{12}^T & A_{22}^T \\ -A_{12}^T & -A_{22}^T \\ -I & 0 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} + \begin{pmatrix} r_1 \\ r_2^+ \\ r_2^- \\ r_3 \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \\ -c_2 \\ 0 \end{pmatrix}, \\
& && r_1 \geq 0,\ r_2^+ \geq 0,\ r_2^- \geq 0,\ r_3 \geq 0,
\end{aligned}$$

where the subvectors of r are named to match the subvectors of x. Equivalently this problem takes the form

$$(D') \qquad \begin{aligned}
&\text{maximize} && b_1^T y_1 + b_2^T y_2 \\
&\text{subject to} && A_{11}^T y_1 + A_{21}^T y_2 + r_1 = c_1, \\
& && A_{12}^T y_1 + A_{22}^T y_2 + r_2^+ = c_2, \\
& && -A_{12}^T y_1 - A_{22}^T y_2 + r_2^- = -c_2, \\
& && -y_1 + r_3 = 0, \\
& && r_1 \geq 0,\ r_2^+ \geq 0,\ r_2^- \geq 0,\ r_3 \geq 0.
\end{aligned}$$

Note that adding the two equalities $A_{12}^T y_1 + A_{22}^T y_2 + r_2^+ = c_2$ and $-A_{12}^T y_1 - A_{22}^T y_2 + r_2^- = -c_2$ gives $r_2^+ + r_2^- = 0$, so that $r_2^+ \geq 0$ and $r_2^- \geq 0$ imply $r_2^+ = r_2^- = 0$, which gives $A_{12}^T y_1 + A_{22}^T y_2 = c_2$. In addition, $-y_1 + r_3 = 0$, $r_3 \geq 0$, is equivalent to $y_1 = r_3 \geq 0$. Therefore, (D') is equivalent to (D), as (D) is the resulting problem where r_1 and r_3 are eliminated.

As for optimality conditions, the primal-dual pair (P_{LP}) and (D_{LP}) give optimality if and only if $\hat{x}$ is feasible to (P_{LP}), $(\hat{y}, \hat{r})$ are feasible to (D_{LP}), and $\hat{x}^T \hat{r} = 0$. For (P') and (D'), complementarity is

$$\hat{x}_1^T \hat{r}_1 + (\hat{x}_2^+)^T \hat{r}_2^+ + (\hat{x}_2^-)^T \hat{r}_2^- + \hat{x}_3^T \hat{r}_3 = 0.$$

Since all four terms are nonnegative, this is equivalent to the four componentwise conditions

$$\hat{x}_1^T \hat{r}_1 = 0, \quad (\hat{x}_2^+)^T \hat{r}_2^+ = 0, \quad (\hat{x}_2^-)^T \hat{r}_2^- = 0, \quad \hat{x}_3^T \hat{r}_3 = 0.$$

As $\hat{r}_2^+ = 0$ and $\hat{r}_2^- = 0$ we immediately obtain $(\hat{x}_2^+)^T \hat{r}_2^+ = 0$ and $(\hat{x}_2^-)^T \hat{r}_2^- = 0$. From (D'), we obtain $\hat{r}_1 = c_1 - A_{11}^T \hat{y}_1 - A_{21}^T \hat{y}_2$, so that $\hat{x}_1^T \hat{r}_1 = 0$ is equivalent to $\hat{x}_1^T (c_1 - A_{11}^T \hat{y}_1 - A_{21}^T \hat{y}_2) = 0$. Finally, (D') gives $\hat{r}_3 = \hat{y}_1$ and (P') gives $\hat{x}_3 = A_{11} \hat{x}_1 + A_{12} (\hat{x}_2^+ - \hat{x}_2^-) - b_1$, so that $\hat{x}_3^T \hat{r}_3 = 0$ is equivalent to $\hat{y}_1^T (A_{11} \hat{x}_1 + A_{12} (\hat{x}_2^+ - \hat{x}_2^-) - b_1) = 0$. As $\hat{x}_2^+ - \hat{x}_2^-$ in (P') corresponds to x_2 in (P), the corresponding complementarity condition in (D) is $\hat{y}_1^T (A_{11} \hat{x}_1 + A_{12} \hat{x}_2 - b_1) = 0$. Thereby, the statements in the formula sheet concerning (P) and (D) have been verified by using (P_{LP}) and (D_{LP}), for which we have derived duality and optimality conditions in the lecture.